Math 515

HW4.

1. (a) Let $a=\hat{t}_0<\hat{t}_1<\cdots<\hat{t}_{n_1}=b$ be the finite time partition associated with ϕ. and $a=\bar{t}_0<\bar{t}_1<\cdots<\bar{t}_{n_2}=b$ with ψ. Let $a=t_0<t_1<\cdots<t_n=b$ be the time partition combining the above two partitions. Then.

$$\int_a^b(\alpha\phi+\beta\psi)\,dW_t=\sum_{j=0}^{n-1}(\alpha\phi(t_j,\omega)+\beta\psi(t_j,\omega))\cdot\Delta W_{t_j}(\omega)$$

$$=\sum_{j=0}^{n-1}\alpha\phi(t_j,\omega)\Delta W_{t_j}(\omega)+\sum_{j=0}^{n-1}\beta\psi(t_j,\omega)\Delta W_{t_j}(\omega)$$

$$=\alpha\sum_{j=0}^{n-1}\phi(t_j,\omega)\Delta W_{t_j}(\omega)+\beta\sum_{j=0}^{n-1}\psi(t_j,\omega)\Delta W_{t_j}(\omega)$$

$$=\alpha\int_a^b\phi\,dW_t+\beta\int_a^b\psi\,dW_t$$

(b). Let $a=t_0<t_1<\cdots<t_{n_1}=b$ be the finite time partition associated with ϕ on $[a,b]$ and $b=t_{n_1}<t_{n_1+1}<\cdots<t_{n_1+n_2}=c$ be the partion associated with ϕ on $[b,c]$.

Then let $n=n_1+n_2$

$$\int_a^b\phi\,dW_t+\int_b^c\phi\,dW_t$$

$$=\sum_{j=0}^{n_1-1}\phi(t_j,\omega)\Delta W_{t_j}(\omega)+\sum_{j=0}^{n-n_1-1}\phi(t_{n_1+j},\omega)\Delta W_{t_{n_1+j}}(\omega)$$

$$=\sum_{j=0}^{n-1}\phi(t_j,\omega)\,\Delta W_{t_j}(\omega)$$

$$=\int_a^c\phi\,dW_t.$$

2. $$\left[\frac{W_t^4}{4}-tW_t\right]_{t=0}^{t=T}=\int_0^T W_t+\frac{1}{2}(3W_t^2)\,dt+\int_0^T(W_t^3-t)\,dW_t.$$

$$\Rightarrow\int_0^T W_t^3-t\,dW_t=\frac{W_T^4}{4}-TW_T-\int_0^T(W_t+\frac{3}{2}W_t^2)\,dt.$$

$$E\left[\frac{W_T^4}{4}-TW_T-\int_0^T(W_t+\frac{3}{2}W_t^2)\,dt\right]$$

$$=\frac{1}{4}E[W_T^4]-T\underbrace{E[W_T]}_{0}-\int_0^T E[W_t+\frac{3}{2}W_t^2]\,dt$$

$$=\frac{3T^2}{4}-\int_0^T\frac{3}{2}t\,dt$$

$$=\frac{3}{4}T^2-\frac{3}{4}T^2=0\rightarrow\text{Martingale properties.}$$

3. (a) $$W_T^2=W_{t_n}^2=W_n^2$$

$$=(W_n^2-W_{n-1}^2)+(W_{n-1}^2-W_{n-2}^2)+\cdots(W_1^2-W_0^2)+W_0^2$$

$$=\sum_{j=0}^{n-1}(W_{j+1}^2-W_j^2)$$

$$\Rightarrow I-I_n=\frac{1}{2}[W_T^2-T]-\sum_{j=0}^{n-1}W_j\Delta W_j$$

$$=\frac{1}{2}W_T^2-\sum_{j=0}^{n-1}W_j(W_{j+1}-W_j)-\frac{1}{2}T$$

$$=\frac{1}{2}\sum_{j=0}^{n-1}(W_{j+1}^2-W_j^2)-\frac{1}{2}\sum_{j=0}^{n-1}2(W_jW_{j+1}-W_j^2)-\frac{1}{2}\sum_{j=0}^{n-1}\Delta t$$

$$=\frac{1}{2}\sum_{j=0}^{n-1}(W_{j+1}^2-2W_jW_{j+1}+W_j^2-\Delta t)$$

$$=\frac{1}{2}\sum_{j=0}^{n-1}[(\Delta W_j)^2-\Delta t]$$

(b) $$E[(I-I_n)^2]=E\left[\frac{1}{2}\sum_{i=0}^{n-1}[(\Delta W_i)^2-\Delta t]\cdot\frac{1}{2}\sum_{j=0}^{n-1}[(\Delta W_j)^2-\Delta t\right.$$

$$=\frac{1}{4}\sum_{i=0}^{n-1}\sum_{j=0}^{n-1}E[((\Delta W_i)^2-\Delta t)((\Delta W_j)^2-\Delta t)]$$

$$=\frac{1}{4}\sum_{i=0}^{n-1}\sum_{j=0}^{n-1}E[(\Delta W_i)^2(\Delta W_j)^2-\Delta t(\Delta W_i)^2-\Delta t(\Delta W_j)^2+(\Delta t)^2]$$

$$=\frac{1}{4}\sum_{i=0}^{n}E[(\Delta W_i)^4-2\Delta t(\Delta W_i)^2+(\Delta t)^2]+\frac{1}{4}\sum_{\substack{i,j=0\\i\neq j}}^{n}E[(\Delta W_i)^2(\Delta W_j)^2-\Delta t(\Delta W_i)^2-\Delta t(\Delta W_j)^2+(\Delta t)^2$$

$$=\frac{1}{4}\sum_{i=0}^{n}\left(E[(\Delta W_i)^4]-2\Delta t E[(\Delta W_i)^2]+(\Delta t)^2\right)+\frac{1}{4}\sum_{\substack{i,j=0\\i\neq j}}^{n-1}\left(E[(\Delta W_i)^2(\Delta W_j)^2-\Delta t E[(\Delta W_i)^2]-\Delta t E[(\Delta W\right.$$
$$+(\Delta t)^2]$$

$$=\frac{1}{4}\sum_{i=0}^{n}(3(\Delta t)^2-2(\Delta t)^2+(\Delta t)^2)+\frac{1}{4}\sum_{\substack{i,j=0\\i\neq j}}^{n-1}((\Delta t)^2-(\Delta t)^2-(\Delta t)^2+(\Delta t)^2)$$

$$=\frac{1}{4}\sum_{i=0}^{n}2(\Delta t)^2=\frac{1}{2}T\Delta t\rightarrow 0$$

as $\Delta t\rightarrow 0$. (i.e. $n\rightarrow+\infty$)

4\. (a) Let $f(t, W_t) = W_t^k$, $k \geq 2$.

Then by the Ito's formula.

$$[W_t^k]_{t=0}^{t=T} = \int_0^t (f_t + \frac{1}{2} f_{ww}) dt + \int_0^t f_w dW_t$$

$$= \int_0^t \frac{1}{2} k(k-1) W_t^{k-2} dt + \int_0^t k W_t^{k-1} dW_t$$

$$\Rightarrow W_T^k = \frac{1}{2} k(k-1) \int_0^t W_t^{k-2} dt + k \int_0^t W_t^{k-1} dW_t$$

$$\Rightarrow E[W_T^k] = \frac{1}{2} k(k-1) \int_0^t E[W_t^{k-2}] dt + k \underbrace{E[\int_0^t W_t^{k-1} dW_t]}_{= 0}$$

$$= \frac{1}{2} k(k-1) \int_0^T E[W_t^{k-2}] dt$$

(b). $$E[W_T^2] = \frac{1}{2} 2 \cdot (2-1) \int_0^T E[W_t^0] dt$$

$$= \int_0^T 1 dt$$

$$= T.$$

$$E[W_T^4] = \frac{1}{2} 4 \cdot (4-1) \int_0^T E[W_t^2] dt$$

$$= 6 \cdot \int_0^T t dt.$$

$$= 3T^2$$

$$E[W_T^6] = \frac{1}{2} \cdot 6 (6-1) \int_0^T E[W_t^4] dt$$

$$= 15 \cdot \int_0^T 3t^2 dt$$

$$= 15 T^3$$